

The Hamiltonian cycle and Hamiltonian paths passing through prescribed edges in a star graph with faulty edges *

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Abstract

Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n-3$, $|E_0| \leq 2n-5-2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n-3$, $|E_0| \leq 2n-7-2|F_e|$.

1 Introduction

The hypercube Q_n and star graph S_n are the most popular and efficient interconnection networks. They possess many excellent properties such as recursive structure, symmetry, small diameter, low degree, popular structure embedding, easy routing, and optimal connectivity and fault-tolerance. There are many literatures for fault tolerance of Hamiltonian properties of hypercube and star graphs.

The interconnection network is usually represented by an undirected graph where vertices represent processors and edges represent links. Let $G = (V_0 \cup V_1, E)$ be a *bipartite graph* where V_0 and V_1 are two disjoint vertex sets such that each edge of E consists of one vertex from each set. A *Hamiltonian path* is a spanning path of G . A *Hamiltonian cycle* is a cycle that visits every vertex exactly once. A *Hamiltonian graph* is a graph that contains a Hamiltonian cycle. Let F_e be a set of faulty edges and $|F_e|$ be the number of F_e . A graph $G = (V, E)$ is *k edges Hamiltonian* if $G - F_e$ is a Hamiltonian graph for every $F_e \subset E$ and $|F_e| \leq k$. A bipartite graph $G = (V_0 \cup V_1, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path* between x, y for any $x \in V_0, y \in V_1$. A bipartite graph $G = (V_0 \cup V_1, E)$ is *k edges Hamiltonian laceable* if $G - F_e$ is Hamiltonian laceable

for every $F_e \subset E$ and $|F_e| \leq k$. In [3], Li et al. proved that S_n is $(n-3)$ edges Hamiltonian laceable and $(n-4)$ edges hyper-Hamiltonian laceable with $n \geq 4$.

On the other hand, Dvořák investigated the problem about prescribed edges. Given a set of fault-free prescribed edges in a hypercube, there is a Hamiltonian cycle passing through all edges of this set in the hypercube. In [1], they showed that Q_n contains a Hamiltonian cycle passing through all edges of any set of prescribed edges E_0 if and only if the subgraph induced by E_0 consists of pairwise vertex-disjoint paths and $|E_0| \leq 2n-3$. Wang et al. further presented the result that all edges of E_0 lie on a Hamiltonian cycle of $Q_n - F_e$ if E_0 is a linear forest (i.e., pairwise vertex-disjoint paths) and $|F_e| \leq n-2$, $|E_0| \leq 2n-3-2|F_e|$ in [4]. Let $d(x, y)$ be the distance of vertices x and y . The authors of [2] introduced that all edges of E_0 lie on a Hamiltonian path $P(x, y)$ of Q_n where E_0 is a linear forest with $|E_0| \leq 2n-4$ and at most one vertex of x, y is incident with one edge of E_0 and $d(x, y)$ is odd.

However, there is no similar results on star graphs. In this paper, we will investigate the problem about prescribed edges of star graphs. We will show that all edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n-3$, $|E_0| \leq 2n-5-2|F_e|$ where E_0 is a linear forest. We also prove that there is a Hamiltonian path $P(x, y)$ passing through all edges in E_0 of $S_n - F_e$ if $|F_e| \leq n-3$ and $|E_0| \leq 2n-7-2|F_e|$ for $n \geq 4$ where E_0 is a linear forest.

2 Preliminaries

In this section, we will introduce some definition, notations and properties about star graphs which are useful for our proofs.

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Definition 1 The n -dimensional star graph is denoted by S_n . The vertex set $V = \{v|v \text{ is a permutation of } 1, 2, \dots, n\}$ and edge set $E = \{(v_1 v_2 \dots v_{i-1} v_i \dots v_n, v_i v_2 \dots v_{i-1} v_1 v_{i+1} \dots v_n) | v_1 \dots v_n \in V \text{ and } 2 \leq i \leq n\}$

A star graph S_n is an edge symmetric and node symmetric graph. By definition, S_n contains $n!$ vertices and $\frac{n!(n-1)}{2}$ edges. The graph S_n can be decomposed to n copies of $(n-1)$ -dimensional star graphs. Give $2 \leq k \leq n$, we can use S_n^i to denote the subgraph of S_n induced by the vertices whose n th bit is i . We further denote the induced subgraph $S_n^{i_1} \cup S_n^{i_2} \dots \cup S_n^{i_k}$ as $S_n^{i_1, i_2, \dots, i_k}$ for $1 \leq i_1, i_2, \dots, i_k \leq n, 1 \leq k \leq n$. Let $\langle n \rangle = \{1, 2, \dots, n\}$ and I_k be the subset of $\langle n \rangle$ with $|I_k| = k$. Let $[u]_k$ be the k th bit of a vertex u .

Lemma 1 There are $(n-2)!$ edges between S_{n-1}^i and S_{n-1}^j where $i \neq j, 1 \leq i \leq n$ and $1 \leq j \leq n$

Lemma 2 If x and $y \in S_n$ and $d(x, y) \leq 2$, then $1(x) \neq 1(y)$.

Li et al. proved the follow lemma in [3].

Lemma 3 The graph S_n is $(n-3)$ -edge Hamiltonian laceable and $(n-4)$ -edge hyper-Hamiltonian laceable with $n \geq 4$.

3 Hamiltonian cycles and paths passing through prescribed edges

In this section, we will show that all edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n-3$, $|E_0| \leq 2n-5-2|F_e|$ and all edges of E_0 lie on a Hamiltonian path $P(x, y)$ of $S_n - F_e$ if $|F_e| \leq n-3$ and $|E_0| \leq 2n-7-2|F_e|$ for $n \geq 4$ where E_0 is a linear forest.

Lemma 4 Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq 1$, $|E_0| \leq 3-2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ of S_n where $d(u, v)$ is odd, $|E_0| = 1$ for $n \geq 4$.

Proof. Suppose that the edge of E_0 is neither incident with u nor v . Since S_n is $(n-3)$ edges Hamiltonian laceable, there is a Hamiltonian path passing through the edge of E_0 . Suppose that the edge of E_0 is incident with one vertex of u and v . Without loss of generality, we can assume

that $E_0 = \{(u, v')\}$. Since S_n hyper-Hamiltonian laceable, there is a Hamiltonian path $P(v', v)$ of $S_n - \{u\}$. Thus, $\langle u, v' \xrightarrow{P(v', v)} v \rangle$ is passing the edge in E_0 of S_n .

In the following, we will prove that all edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$. We will first prove the case for $n = 4$. Suppose that $F_e = \emptyset$ and $|E_0| = 3$. There is a Hamiltonian cycle of Table 1 which can pass through all edges of E_0 . Suppose that $|F_e| = 1$ and $|E_0| = 1$. Since S_n is edge transitive, we can assume that $F_e = \{(2413, 3412)\}$. There is a Hamiltonian cycle of Table 2 which can pass through the edge of E_0 .

Next, we will prove the case for $n \geq 5$. By the symmetry of S_n , we can arrange every edge of $E_0 \cup F_e$ is in some sub-star graph. Without loss of generality, we can assume that S_n^1 has the most number of E_0 . We can construct a Hamiltonian cycle $\langle x, y \xrightarrow{P(y, x)} x \rangle$ passing through every prescribed edge of $S_n^1 - F_e^1$. Since there is a Hamiltonian path between any two vertices with odd distance, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ passing through every prescribed edge in $S_n^{2, \dots, n}$ of $S_n^{2, \dots, n} - F_e$. Thus, $\langle x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y, x)} x \rangle$ is the Hamiltonian cycle passing through every edge of E_0 of $S_n - F_e$. $\square$

Lemma 5 Let $I = \{2, 3, \dots, n\}$. Let s and t be arbitrary vertices of $S_n^I = (V_0 \cup V_1, E)$ and $e \in E$ be a prescribed edge for $s \in V_0, t \in V_1, e \neq (s, t)$ and $n \geq 5$. There exists a Hamiltonian path $P(s, t)$ passing through the edge e of S_n^I .

Proof. The proof is divided into three cases:

Case 1: $s, t \in S_n^i$ for some $2 \leq i \leq n$.

Without loss of generality, we can assume that $i = 2$. Applying Lemma 4, we can construct a Hamiltonian path $\langle s \xrightarrow{P(s, x)} x, y \xrightarrow{P(y, t)} t \rangle$ passing through a prescribed edge of S_n^2 with $[x]_1, [y]_1 \neq 1$. Applying Lemma 4, we can construct a Hamiltonian path between any two vertices with odd distance passing through a prescribed edge in S_n^j for $3 \leq j \leq n$. Thus, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ passing through a prescribed edge of $S_n^{3, \dots, n}$. Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y, t)} t \rangle$ is the Hamiltonian path passing through the edge e of S_n^I .

Case 3.4 $k_1 = 2, k_2 = n$ and $u \in V_0$.

Let $\phi(x)$ be a neighbor of $\phi(u)$ in S_n^n with $x \in S_n^{n-1}$. Applying Lemma 3, we can construct a Hamiltonian path $\langle t \xrightarrow{P(t, \phi(y))} \phi(y), \phi(u), \phi(x) \rangle$ of S_n^n for $y \in S_n^3$. Let z be a neighbor of u in S_n^2 with $\phi(z) \in S_n^3$. Applying Lemma 3, we can construct a Hamiltonian path $\langle s \xrightarrow{P(s, a)} a, u, z \rangle$ of S_n^2 for $\phi(a) \in S_n^4$. We can also construct the Hamiltonian paths $P(\phi(z), y)$ of S_n^3 and $P(\phi(a), x)$ of $S_n^{4, \dots, n-1}$. Thus, $\langle s \xrightarrow{P(s, a)} a, \phi(a) \xrightarrow{P(\phi(a), x)} x, \phi(x), \phi(u), u, z, \phi(z) \xrightarrow{P(\phi(z), y)} y, \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ is the Hamiltonian path passing through the edge e of S_n^I , as illustrated in Figure 2.

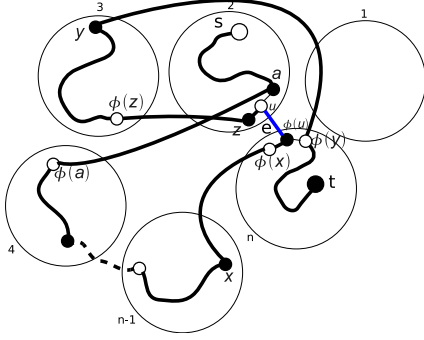


Figure 2: The illustration of case 3.4 of Lemma 5.

Case 3.5 $k_1 = 2, k_2 = n$ and $u \in V_1$.

Applying Lemma 3, we can construct a Hamiltonian path $\langle s \xrightarrow{P(s, u)} u \rangle$ of S_n^2 and $\langle \phi(u) \xrightarrow{P(\phi(u), \phi(x))} \phi(x), \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ of S_n^n for $x \in S_n^3$ and $y \in S_n^{n-1}$. We can also construct the Hamiltonian path $\langle x \xrightarrow{P(x, y)} y \rangle$ of $S_n^{3, \dots, n-1}$. Thus, $\langle s \xrightarrow{P(s, u)} u, \phi(u) \xrightarrow{P(\phi(u), \phi(x))} \phi(x), x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ is the Hamiltonian path passing through e of S_n^I . $\square$

Lemma 6 Let $I = \{2, 3, \dots, n\}$. Let s_1, s_2 and t_1, t_2 be vertices of $S_n^I = (V_0 \cup V_1, E)$ for $s_1, s_2 \in V_0, t_1, t_2 \in V_1, [s_1]_1 = [s_2]_1 = [t_1]_1 = [t_2]_1 = 1$ $n \geq 5$. There exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of S_n^I .

Proof. Let $U = \{s_1, s_2, t_1, t_2\}$ and $U^i = U \cap V(S_n^i)$ for $2 \leq i \leq n$. The proof is divided into cases:

Case 1: $|U^i| = 4$ for some $2 \leq i \leq n$.

Without loss of generality, we can assume that $i = n$. Applying Lemma 3, we can construct a Hamiltonian path $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), s_1 \xrightarrow{P(s_1, t_1)} t_1, \phi(y) \xrightarrow{P(\phi(y), t_2)} t_2 \rangle$ of S_n^n . Suppose that $x, y \in S_n^j$ for some $2 \leq j \leq n-1$. Without loss of generality, we can assume that $j = 2$. Applying Lemma 3, we can construct a Hamiltonian path $\langle x \xrightarrow{P(x, a)} a, b \xrightarrow{P(b, y)} y, \phi(b) \in S_n^{k_1}$ and $\phi(a) \in S_n^{k_2}$ for $3 \leq k_1, k_2 \leq n-1$. Without loss of generality, we can assume that $k_1 = 3$ and $k_2 = n-1$. We can also construct the Hamiltonian path $P(\phi(b), \phi(a))$ of $S_n^{3, \dots, n-1}$. Thus, $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), x \xrightarrow{P(x, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, y)} y, \phi(y) \xrightarrow{P(\phi(y), t_2)} t_2 \rangle$ and $P(s_1, t_1)$ are two spanning paths of S_n^I , as illustrated in Figure 3.

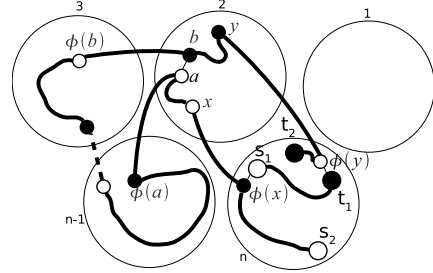


Figure 3: The illustration of case 1 of Lemma 6.

Suppose that $x \in S_n^i, y \in S_n^j, 2 \leq i \neq j \leq n-1$. Without loss of generality, we can assume that $i = n-1$ and $j = 2$. We can construct a Hamiltonian path $P(\phi(b), \phi(a))$ of $S_n^{2, 3, \dots, n-1}$. Thus, $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), t_2)} t_2 \rangle$ and $P(s_1, t_1)$ are two spanning paths of S_n^I .

Case 2 $|U^i| = 3$ and $|U^j| = 1$ for some $2 \leq i \neq j \leq n$.

Without loss of generality, we can assume that $s_1, t_1, t_2 \in S_n^n$ and $s_2 \in S_n^2$. Applying Lemma 3, we can construct a Hamiltonian path $\langle s_1 \xrightarrow{P(s_1, t_1)} t_1, \phi(x) \xrightarrow{P(\phi(x), t_2)} t_2 \rangle$ of S_n^n with $x \notin S_n^2$. Without loss of generality, we can assume that $x \in S_n^{n-1}$. Applying Lemma 3, we can construct Hamiltonian path $P(s_2, y)$ of S_n^2 for $\phi(y) \in S_n^3$. We can also construct the Hamiltonian path $\langle \phi(y) \xrightarrow{P(\phi(y), x)} x \rangle$ of $S_n^{3, \dots, n-1}$. Thus, $\langle s_2 \xrightarrow{P(s_2, y)} y, \phi(y) \xrightarrow{P(\phi(y), x)} x \rangle$

$x, \phi(x) \xrightarrow{P(\phi(x), t_2)} t_2$ and $\langle s_1 \xrightarrow{P(s_1, t_1)} t_1 \rangle$ are two spanning paths of S_n^I .

Case 3 $|U^i| = 2$ and $|U^j| = 2$ for some $2 \leq i \neq j \leq n$.

Without loss of generality, we can assume that $i = 2$ and $j = n$.

Case 3.1 $s_1, t_2 \in S_n^2, s_2, t_1 \in S_n^n$.

Let x be a neighbor of t_2 in S_n^2 with $\phi(x) \in S_n^n$. Applying Lemma 3, we can construct Hamiltonian path $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), \phi(y) \xrightarrow{P(\phi(y), t_1)} t_1 \rangle$ of S_n^n for $y \notin S_n^1$. Without loss of generality, we can assume that $y \in S_n^{n-1}$. Applying Lemma 3, we can construct Hamiltonian path $P(s_1, z), x, t_2$ of S_n^2 for $\phi(z) \in S_n^3$. We can also construct the Hamiltonian path $P(\phi(z), y)$ of $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, z)} z, \phi(z) \xrightarrow{P(\phi(z), y)} y, \phi(y) \xrightarrow{P(\phi(y), t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), x, t_2 \rangle$ are two spanning paths of S_n^I .

Case 3.2 $s_1, s_2 \in S_n^n, t_1, t_2 \in S_n^2$.

Let x be a neighbor of t_1 in S_n^2 with $\phi(x) \in S_n^n$. Applying Lemma 3, we can construct Hamiltonian path $\langle s_2 \xrightarrow{P(s_2, \phi(y))} \phi(y), s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x) \rangle$ of S_n^n for $y \notin S_n^{2,1}$. Without loss of generality, we can assume that $y \in S_n^{n-1}$. Applying Lemma 3, we can construct Hamiltonian path $\langle x, t_1, z \xrightarrow{P(z, t_2)} t_2 \rangle$ of S_n^2 for $\phi(z) \in S_n^3$. We can also construct the Hamiltonian path $P(y, \phi(z))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x), x, t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, \phi(y))} \phi(y), y \xrightarrow{P(y, \phi(z))} \phi(z), z \xrightarrow{P(z, t_2)} t_2 \rangle$ are two spanning paths of S_n^I .

Case 3.3 $s_1, t_1 \in S_n^2, s_2, t_2 \in S_n^n$.

Applying Lemma 3, we can construct Hamiltonian path $\langle s_1 \xrightarrow{P(s_1, x)} x, y \xrightarrow{P(y, t_1)} t_1 \rangle$ of S_n^2 . Applying Lemma 3, also $P(s_2, t_2)$ of S_n^n . Suppose that $\phi(x) \in S_n^i, \phi(y) \in S_n^j, 3 \leq i \neq j \leq n-1$. Without loss of generality, we can assume that $i = 3$ and $j = n-1$. We can also construct the Hamiltonian path $P(\phi(x), \phi(y))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y, t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, t_2)} t_2 \rangle$ are two spanning paths of S_n^I .

Case 4 $|U^i| = 2, |U^j| = 1$ and $|U^k| = 1$ for some different integers $2 \leq i, j, k \leq n$.

Without loss of generality, we can assume that $i = 2, j = 3$ and $k = n$. We can also assume that $t_2 \in S_n^2$.

Case 4.1 $s_1, t_2 \in S_n^2$.

Let x be a neighbor of t_2 in S_n^2 with $\phi(x) \in S_n^n$. We can also construct the adjacent vertices x of t_2 of S_n^2 for $\phi(x) \in S_n^n$. Applying Lemma 3, we can construct Hamiltonian path $P(s_2, \phi(x))$. Applying Lemma 3, we can construct Hamiltonian path $\langle s_1 \xrightarrow{P(s_1, y)} y, x, t_2 \rangle$ of S_n^2 for $\phi(y) \notin S_n^{2,1}$. Without loss of generality, we can assume that $\phi(y) \in S_n^{n-1}$. We can also construct the Hamiltonian path $\langle \phi(y) \xrightarrow{P(\phi(y), t_1)} t_1 \rangle$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, y)} y, \phi(y) \xrightarrow{P(\phi(y), t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), x, t_2 \rangle$ are two spanning paths of S_n^I .

Case 4.2 $t_1, t_2 \in S_n^2$.

Let y be a neighbor of t_2 in S_n^2 with $\phi(y) \in S_n^n$. Applying Lemma 3, we can construct Hamiltonian path $P(s_2, \phi(y))$ of S_n^n . Applying Lemma 3, we can construct Hamiltonian path $\langle y, t_2, x \xrightarrow{P(x, t_1)} t_1 \rangle$ of S_n^2 for $\phi(x) \notin S_n^{2,1}$. Without loss of generality, we can assume that $\phi(x) \in S_n^{n-1}$. We can also construct the Hamiltonian path $P(s_1, \phi(x))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x), x \xrightarrow{P(x, t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, \phi(y))} \phi(y), y, t_2 \rangle$ are two spanning paths of S_n^I .

Case 4.3 $s_2, t_2 \in S_n^2$.

Without loss of generality, we can assume that $t_1 \in S_n^3, s_1 \in S_n^n$. Applying Lemma 3, we can construct Hamiltonian path $P(s_2, t_2)$ of S_n^2 . Suppose that $x \in S_n^i, 2 \leq i \leq n-1$. Applying Lemma 3, we can construct Hamiltonian path $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x) \rangle$ of S_n^n , we can also assume that $x \in S_n^{n-1}$. We can also construct the Hamiltonian path $P(x, t_1)$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x), x \xrightarrow{P(x, t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, t_2)} t_2 \rangle$ are two spanning paths of S_n^I .

Case 5 The four vertices s_1, s_2, t_1, t_2 are in different subgraphs.

Without loss of generality, we can assume

that $s_1 \in S_n^3$, $s_2 \in S_n^2$, $t_2 \in S_n^{n-1}$ and $t_1 \in S_n^n$. Applying Lemma 3, we can construct Hamiltonian path $\langle s_1 \xrightarrow{P(s_1, t_1)} t_1 \rangle$ of $S_n^{2,n}$. Applying Lemma 3, we can construct Hamiltonian paths $\langle s_2 \xrightarrow{P(s_2, t_2)} t_2 \rangle$ of $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, t_2)} t_2 \rangle$ are two spanning paths of S_n^I . $\square$

Let $E'(S_n^i)$ be the set of edges incident to at least one vertex of S_n^i . That is, $E'(S_n^i) = E(S_n^i) \cup \{(u, v) | u \in V(S_n^i), v \in S_n^j \text{ for } j \neq i\}$. Let F_e^i be the set of faulty edges in $E'(S_n^i)$ and E_0^i be the prescribed edges in $E'(S_n^i)$ for $1 \leq i \leq n$.

Theorem 1 *Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n - 3$, $|E_0| \leq 2n - 5 - 2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n - 3$, $|E_0| \leq 2n - 7 - 2|F_e|$.*

Proof. We will prove this theorem by induction on n . By symmetry of star graph, we can arrange every faulty edge of F_e in some S_n^i and at most one prescribed edge of E_0 is not in any S_n^i for $1 \leq i \leq n$. We first show that all edge of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n - 3$, $|E_0| \leq 2n - 5 - 2|F_e|$ with the following cases.

Case 1: $2|F_e^k| + |E_0^k| = 2n - 5$ for some $1 \leq k \leq n$.

Without loss of generality, we can assume that $k = 1$. By induction hypothesis, we can construct a Hamiltonian cycle passing through any prescribed edges E_0' of $S_n^1 - F_e'$ with $2|F_e'| + |E_0'| \leq 2n - 7$. Thus, we can construct two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ passing through E_0 of $S_n^1 - F_e$. Without loss of genrality, we can assume that $s_1, s_2 \in V_1$ and $t_1, t_2 \in V_0$. Applying Lemma 6, we can construct two spanning paths $P(\phi(t_2), \phi(s_1))$ and $P(\phi(t_1), \phi(s_2))$ in $S_n^{2,3, \dots, n}$. Thus, $\langle s_1 \xrightarrow{P(s_1, t_1)} t_1, \phi(t_1) \xrightarrow{P(\phi(t_1), \phi(s_2))} \phi(s_2), s_2 \xrightarrow{P(s_2, t_2)} t_2, \phi(t_2) \xrightarrow{P(\phi(t_2), \phi(s_1))} \phi(s_1), s_1 \rangle$ is the Hamiltonian cycle passing through E_0 of $S_n - F_e$.

Case 2: $2|F_e^k| + |E_0^k| = 2n - 6$ for some $1 \leq k \leq n$.

Without loss of generality, we can assume that $k = 1$. By induction hypothesis, we can construct a Hamiltonian cycle passing through any prescribed edges E_0' of $S_n^1 - F_e'$

with $2|F_e'| + |E_0'| \leq 2n - 7$. Thus, we can construct a Hamiltonian path $P(s, t)$ passing through E_0' of $S_n^1 - F_e'$. Let $\{e_0\} = E_0 - E_0'$. Applying Lemma 5, we can construct a Hamiltonian path $P(\phi(t), \phi(s))$ passing through e_0 of $S_n^{2,3, \dots, n}$. Thus, $\langle s \xrightarrow{P(s, t)} t, \phi(t) \xrightarrow{P(\phi(t), \phi(s))} \phi(s), s \rangle$ is the Hamiltonian cycle passing through E_0 of $S_n - F_e$.

Case 3: $2|F_e^i| + |E_0^i| \leq 2n - 7$ for every $1 \leq k \leq n$.

Without loss of generality, we can assume that $2|F_e^i| + |E_0^i| \geq 2|F_e^j| + |E_0^j|$ for $1 \leq i \leq j \leq n$. Let $(x, \phi(x)), (y, \phi(y)), (z, \phi(z))$ be fault-free edges where $x, y \in S_n^1, \phi(y), z \in S_n^2, \phi(x) \in S_n^3, \phi(z) \in S_n^n$ and $x \in V_1, y, z \in V_0$. By induction hypothesis, we can construct Hamiltonian cycles $\langle x \xrightarrow{P(x, y)} y, x \rangle$ passing through E_0^1 of $S_n^1 - F_e^1$ and $\langle \phi(y) \xrightarrow{P(\phi(y), z)} z, \phi(y) \rangle$ passing through E_0^2 of $S_n^2 - F_e^2$. There exists a Hamiltonian path passing through E_0^i of $S_n^i - F_e^i$ between any two vertices with different color for $3 \leq i \leq n$. Thus, we can construct a Hamiltonian path $P(\phi(z), \phi(x))$ passing through $E_0 - E_0^1 - E_0^2$ of $S_n^{3, \dots, n} - (F_e - F_e^1 - F_e^2)$. Thus, $\langle x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), z)} z, \phi(z) \xrightarrow{P(\phi(z), \phi(x))} \phi(x), x \rangle$ is the Hamiltonian cycle passing through E_0 of $S_n - F_e$.

In the following, we will show that all edges of E_0 lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n - 3$, $|E_0| \leq 2n - 7 - 2|F_e|$, for any $u \in V_1, v \in V_0$.

Case 1: $2|F_e^k| + |E_0^k| = 2n - 7$ for some $1 \leq k \leq n$.

Without loss of generality, we can assume that $k = 1$. By induction hypothesis, there exists a Hamiltonian cycle C passing through E_0 of $S_n^1 - F_e$.

Case 1.1 $u, v \in S_n^1$.

Without loss of generality, we can assume that $C = \langle u \xrightarrow{P(u, x)} x, a \xrightarrow{P(a, v)} v, c \xrightarrow{P(c, b)} b, u \rangle$. Applying Lemma 6, we can construct two spanning disjoint paths $P(\phi(x), \phi(b))$ and $P(\phi(c), \phi(a))$ of $S_n^{2,3, \dots, n}$. Thus, $\langle u \xrightarrow{P(u, x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(b))} \phi(b), b \xrightarrow{P(b, c)} c, \phi(c) \xrightarrow{P(\phi(c), \phi(a))} \phi(a), a \xrightarrow{P(a, v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 1.2 One of u, v in S_n^k the other in S_n^i for some $1 \leq i \neq k \leq n$.

Without loss of generality, we can assume

that $v \in S_n^n$ and $C = \langle u \xrightarrow{P(u,x)} x, u \rangle$ with $(u, x) \notin E_0$. Applying Lemma 5, we can construct a Hamiltonian path $P(\phi(x), v)$ in $S_n^{2,3,\dots,n}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 1.3 $u, v \in S_n^i$ for some $1 \leq i \neq k \leq n$.

Without loss of generality, we can assume that $i = n$ and $C = \langle x \xrightarrow{P(x,y)} y, x \rangle$ with $x \in V_1, (x, y) \notin E_0, [y]_1 = n, [x]_1 = 2, \phi(y) \neq u$. Applying Lemma 3, we can construct a Hamiltonian path $\langle u \xrightarrow{P(u,z)} z, \phi(y) \xrightarrow{P(\phi(y),v)} v \rangle$ of S_n^n with $[z]_1 = n - 1$. We can also construct a Hamiltonian path $P(\phi(z), \phi(x))$ of $S_n^{3,\dots,n}$. Thus, $\langle u \xrightarrow{P(u,z)} z, \phi(z) \xrightarrow{P(\phi(z),\phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y),v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 4

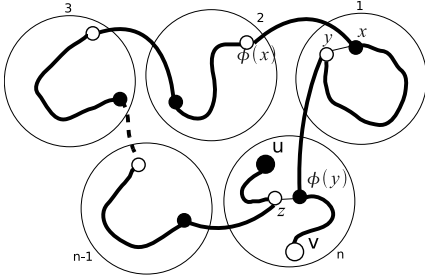


Figure 4: The illustration of case 1.3 of Theorem 1.

Case 1.4 $u \in S_n^i, v \in S_n^j$ for some disjoint integer $1 \leq i, j, k \leq n$.

Without loss of generality, we can assume that $i = 2, j = n$ and $C = \langle x \xrightarrow{P(x,y)} y, x \rangle$ with $x \in V_1, (x, y) \notin E_0, [y]_1 = n, [x]_1 = 3$. Applying Lemma 3, we can construct a Hamiltonian path $P(\phi(y), v)$ of S_n^n . We can also construct a Hamiltonian path $P(u, \phi(x))$ of $S_n^{2,\dots,n-1}$. Thus, $\langle u \xrightarrow{P(u,\phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y),v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2: $2|F_e^k| + |E_0^k| = 2n - 8$ for some $1 \leq k \leq n$.

Without loss of generality, we can assume that $k = 1$ and $[u]_1 \leq [v]_1$.

Case 2.1 $u, v \in S_n^1$.

By induction hypothesis, we can construct

a Hamiltonian path between u and v passing through $2n - 9 - 2|F_e|$ edges of $S_n^1 - F_e$. Thus, we can construct two spanning disjoint paths $P(u, x)$ and $P(y, v)$ passing through E_0 of $S_n^1 - F_e$ where $d(x, y)$ is odd. Applying Lemma 5, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ in $S_n^{2,3,\dots,n}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2.2 $u \in S_n^1$ and $v \in S_n^i$ for some $2 \leq i \leq n$.

Without loss of generality, we can assume that $i = n$. Let $e_0 \in E_0^j$ for some $j \neq 1$. By induction hypothesis, we can construct a Hamiltonian cycle $\langle u \xrightarrow{P(u,x)} x, u \rangle$ passing through E_0^1 of $S_n^1 - F_e^1$. Applying Lemma 5, we can construct a Hamiltonian path $P(\phi(x), v)$ passing through e_0 in $S_n^{2,3,\dots,n}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2.3 $u, v \in S_n^i$ for $i \neq 1$.

Without loss of generality, we can assume that $i = n$. By induction hypothesis, we can construct a Hamiltonian path $\langle u \xrightarrow{P(u,x)} x, y \xrightarrow{P(y,v)} v \rangle$ passing through a prescribed edge of S_n^n where $[x]_1 = 1$ and there is not prescribed edge incident $\phi(x)$. By induction hypothesis, we can construct a Hamiltonian cycle $\langle \phi(x) \xrightarrow{P(\phi(x),z)} z, \phi(x) \rangle$ passing through E_0^1 of $S_n^1 - F_e^1$ with $[z]_1 \neq [y]_1$. There exists a Hamiltonian path between any two vertices with odd distance passing through a prescribed edge in S_n^j for $2 \leq j \leq n - 1$. Thus, we can construct a Hamiltonian path $P(\phi(z), \phi(y))$ passing through a prescribed edge of $S_n^{2,\dots,n-1}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),z)} z, \phi(z) \xrightarrow{P(\phi(z),\phi(y))} \phi(y), y \xrightarrow{P(y,v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 5.

Case 2.4 $u \in S_n^i, v \in S_n^j$ for $2 \leq i \leq j \leq n$.

Without loss of generality, we can assume that $i = 2$ and $j = n$. Let $e_0 = (y, y')$ be the prescribed edge who is not in $E'(S_n^1)$ for $y \in V_1; y' \in V_0$.

Case 2.4.1 $u, y \in S_n^2$ and $v, y' \in S_n^n$.

Let b be a black vertex of S_n^1 where there is no prescribed edge incident to

$\langle a, b \xrightarrow{P(b,a)} a \rangle$ passing through E_0^1 of $S_n^1 - F_e^1$ with $\phi(a) \in S_n^{n-1}$. Applying Lemma 3, we can construct Hamiltonian path $P(\phi(a), y)$ of S_n^{n-1} and a Hamiltonian path $P(u, x)$ of $S_n^{2, \dots, n-2}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(b))} \phi(b), b \xrightarrow{P(b,a)} a, \phi(a) \xrightarrow{P(\phi(a), y)} y, y' \xrightarrow{P(y', v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 8.

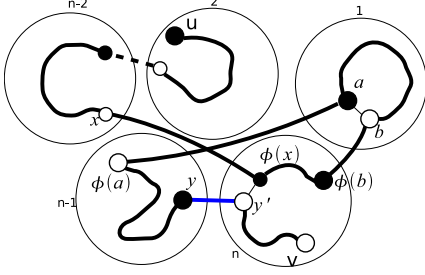


Figure 8: The illustration of case 2.4.3 of Theorem 1.

Case 2.4.4 $[y]_n = [y']_n$ or $(y \notin S_n^2 \text{ and } y' \notin S_n^2 \text{ and } \{[y]_n, [y']_n\} \neq \{2, n\})$.

Without loss of generality, we can assume that $e_0 \notin (E'(S_n^2) - E(S_n^2))$. By induction hypothesis, we can construct a Hamiltonian path $P(u, \phi(a))$ passing through a prescribed edge in S_n^2 where $a \in S_n^1$ and there is no prescribed edge incident to a . By induction hypothesis, we can construct a Hamiltonian cycle $\langle a \xrightarrow{P(a,b)} b, a \rangle$ passing through E_0^1 of $S_n^1 - F_e^1$ with $\phi(b) \in S_n^3$. By induction hypothesis, we can construct a Hamiltonian path $P(\phi(b), v)$ passing through a prescribed edge in $S_n^{3, \dots, n}$. Thus, $\langle u \xrightarrow{P(u, \phi(a))} \phi(a), a \xrightarrow{P(a,b)} b, \phi(b) \xrightarrow{P(\phi(b), v)} v \rangle$.

Case 3: $2|F_e^i| + |E_0^i| \leq 2n - 9$ for every $1 \leq k \leq n$.

Case 3.1 $u, v \in S_n^i$ for some $1 \leq i \leq n$.

Without loss of generality, we can assume that $i = n$. By induction hypothesis, we can construct a Hamiltonian path $\langle u \xrightarrow{P(u,x)} x, y \xrightarrow{P(y,v)} v \rangle$ passing through E_0^n of $S_n^n - F_e^n$. By induction hypothesis, there exists a Hamiltonian path between any two vertices with odd distance passing through E_0^j

in $S_n^j - F_e^j$ for $1 \leq j \leq n-1$. Thus, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ passing through $E_0 - E_0^n$ of $S_n^{1,2, \dots, n-1} - (F_e - F_e^n)$. Therefore, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y,v)} v \rangle$ is the Hamiltonian path passing through $|E_0|$ of $S_n - F_e$.

Case 3.2 $u \in S_n^i, v \in S_n^j$ for some $1 \leq i \neq j \leq n$. By induction hypothesis, there exists a Hamiltonian path between any two vertices with odd distance passing through E_0^j in $S_n^j - F_e^j$ for $1 \leq j \leq n$. Thus, we can construct a Hamiltonian path $P(u, v)$ passing through E_0 of $S_n - F_e$. $\square$

4 Conclusion

In this paper, we show for $n \geq 5$ let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n - 3$, $|E_0| \leq 2n - 5 - 2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n - 3$, $|E_0| \leq 2n - 7 - 2|F_e|$.

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